מבוא למדעי המחשב - הרצאה 3

הערה

סיכום קצר

- **הנושא המרכזי:**
- **הרעיון המרכזי / מה לקחת מההרצאה:**

1. מושגים והגדרות מרכזיים

חשוב

הגדרות ותכונות נשמרות

2. סיכום ההרצאה

נושא 1

הערה

הערות והסברים

y = ++x changes both variables

y = x++ changes only y and not the x variable.

```
((y=2) == x) // (y=2) returns 2
```

Logical operators

|| logical OR

&& logical AND

! logical NOT

if something isnt 0 (false) it will be 1 (true)

on an OR statement, if the first half of the statement is false, it wont check the second half and will just skip it,
so if we went the second half to be accomplished then we need to put it on the first half.
for example:

```
int x=2;
int y =3;
(x&&(y=7));
```

y=7 will not happen since x is already true, if we wanted y=7 to happen we would first need to put it in the first half.

Expressions

a combination of operators and operands. or just an operand.
each expression has:

- type
- value
- address
  a type and value are derived the operand's type and value

RULE: the operands in an expression should be of the same type

either we can make it happen (explicit converion) or the compiler will make it happen even if we don't ask (implicit conversion).

implicit conversion

invoked automatically by the compiler.
rule of thumb: expression converts to higher rank.

```
double d = 2.5;
int a=3,b=4
```

```
a=b+d;
```

b converted from int to double (promotion)

b +d computed as double

a is assigned the result, converting double to int.

Explicit Conversion

invoked by the programmer's code, using (type)

```
double d = 7.5;
int a = (int)d; // a is now 7
int b = 2;
int c = (double)a/b; // 7.0 / 2.0 =3.5 yet c is int so result would be 3
```

every demotion causes loss in data (use with caution).

Notes

the original's variable type doesn't change even though its temporal state does.

%d reads the float's bits as if they were an int

indentation does not matter, just like in java\C#.

Nested conditions

```
if(statement)
        if(statement)
                code
```

Switch Case

only accepts an integer (or char) as a condition, you can cast a float into an integer so the switch case would work.

infinite notes

most times its a big no-no.
unless it is necessary to have an infinite loop on a specific program, its your responsibility to make sure the loop isnt infinite

do-while

instead of being stuck in this:

```
printf(...);
scanf(...);
while(condition){
printf(...);
scanf(...);
}
```

we can use do while to do this instead:

```
do{
printf(...);
scanf(...);
}while(condition)
```